

SALLA NARUSKA, Finland

WMO: 027450

Lat: 67.162N

Lon: 29.178E

Elev: 215

StdP: 98.77

Time Zone: 2.00 (EUE)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.2	-31.6	-37.8	0.1	-34.3	-34.7	0.1	-31.6	6.3	-4.2	5.4	-4.9	0.3	0	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	25.2	16.5	23.1	15.7	21.0	15.1	18.3	23.1	16.9	21.5	15.7	19.3	2.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.6	12.1	20.1	15.2	11.1	18.1	14.1	10.3	17.3	52.5	22.8	48.3	21.5	44.8	19.3	21.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.5	4.9	4.3	-38.8	27.2	2.6	2.7	-40.7	29.1	-42.2	30.7	-43.6	32.3	-45.5	34.2		
(5)			-38.3	20.1	3.0	1.6	-40.5	21.2	-42.2	22.2	-43.9	23.1	-46.0	24.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.5	-14.4	-12.6	-8.9	-1.6	5.5	10.0	13.6	10.9	6.7	-0.6	-5.6	-10.2		
	DBStd	10.95	7.66	8.90	6.44	4.77	4.48	3.99	3.47	3.34	3.20	4.56	5.95	7.53		
	HDD10.0	4087	757	633	587	347	152	48	9	30	104	327	468	625		
	HDD18.3	6889	1015	867	845	597	398	251	151	231	348	586	718	883		
	CDD10.0	246	0	0	0	0	13	48	119	58	7	0	0	0		
	CDD18.3	6	0	0	0	0	0	2	3	1	0	0	0	0		
	CDH23.3	166	0	0	0	0	9	41	93	22	0	0	0	0		
	CDH26.7	24	0	0	0	0	0	6	14	4	0	0	0	0		
	WSAvg	1.8	1.2	1.5	2.0	2.0	2.3	2.3	1.9	1.8	1.8	1.8	1.5	1.5		
(15) Precipitation	PrecAvg	584	38	33	30	32	46	65	80	66	57	54	42	40		
	PrecMax	783	71	75	57	74	70	119	128	134	98	115	76	74		
	PrecMin	449	14	13	8	13	23	18	30	18	13	21	12	11		
	PrecStd	80	14	16	14	14	13	29	26	29	24	23	16	13		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.1	2.8	5.7	13.8	24.6	27.2	28.7	26.4	18.5	10.1	5.2	1.8		
		MCWB	-0.5	1.7	2.6	7.3	13.0	16.9	19.0	19.2	13.3	9.3	4.9	1.3		
	2%	DB	-1.3	1.0	3.0	8.7	21.6	24.1	25.9	22.5	16.2	8.1	3.2	0.7		
		MCWB	-1.6	0.4	0.5	4.1	12.4	16.0	17.2	15.8	12.1	6.8	2.7	0.4		
	5%	DB	-2.7	0.1	1.5	6.7	17.5	20.8	23.8	20.3	14.6	6.8	1.8	0.0		
		MCWB	-3.1	-0.5	-0.2	2.8	11.0	14.1	16.2	15.4	11.5	5.8	1.5	-0.2		
	10%	DB	-4.2	-1.3	0.4	5.3	14.3	18.2	21.8	18.2	12.9	5.4	1.0	-1.0		
		MCWB	-4.5	-1.8	-0.8	1.9	8.7	12.2	15.9	14.1	10.8	4.7	0.7	-1.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.1	1.8	2.7	7.4	14.2	18.4	20.1	19.9	14.7	9.4	5.0	1.5		
		MCDB	0.2	2.6	5.2	13.5	21.7	25.2	24.6	25.6	16.8	10.0	5.1	1.7		
	2%	WB	-1.5	0.6	1.1	4.5	12.9	16.9	18.8	17.2	13.0	7.0	2.7	0.5		
		MCDB	-1.3	0.9	2.4	7.8	20.5	22.9	23.2	21.0	15.1	7.7	3.0	0.7		
	5%	WB	-3.1	-0.4	0.1	3.2	11.3	14.8	17.7	15.7	12.0	5.8	1.6	-0.2		
		MCDB	-2.7	0.1	1.1	5.8	17.5	18.4	22.1	18.9	13.8	6.5	1.8	-0.1		
	10%	WB	-4.5	-1.8	-0.7	2.5	9.2	13.2	16.5	14.7	10.9	4.8	0.7	-1.3		
		MCDB	-4.1	-1.3	0.3	4.6	13.3	17.1	20.5	17.5	12.6	5.4	0.9	-1.1		
(35) Mean Daily Temperature Range	MDBR		9.3	9.3	13.5	11.1	10.8	10.7	11.6	11.4	8.9	5.6	5.5	7.7		
	5% DB	MCDBR	8.6	7.0	11.2	12.1	17.8	16.1	16.7	15.8	13.5	6.4	4.6	5.4		
		MCWBR	8.6	6.7	9.3	7.9	9.6	8.2	9.0	9.9	9.5	5.5	4.5	5.4		
	5% WB	MCDBR	8.7	6.9	8.0	9.8	17.0	13.3	14.3	13.4	11.5	5.6	4.4	5.3		
		MCWBR	8.7	6.6	6.9	6.5	9.6	7.3	8.3	8.9	8.5	5.0	4.4	5.2		
(40) Clear-Sky Solar Irradiance	taub		0.193	0.256	0.293	0.323	0.341	0.342	0.357	0.337	0.310	0.277	0.353	N/A		
	taud		1.920	2.089	2.145	2.182	2.385	2.466	2.445	2.474	2.530	2.321	2.721	N/A		
	Ebn at Noon		161	563	746	825	855	865	836	817	757	556	194	0		
	Edh at Noon		13	51	84	107	99	94	93	81	60	41	15	0		
(44) All-Sky Solar Radiation	RadAvg		0.06	0.55	2.01	3.76	4.80	5.01	4.60	3.32	1.76	0.57	0.12	0.00		
	RadStd		0.00	0.08	0.17	0.28	0.47	0.40	0.49	0.32	0.17	0.10	0.02	0.00		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.60	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page